

SPY V17 Conservative Model

Treasury Live-Test Report and Operating Runbook

Prepared after code/data audit. Cached data tested through **2026-04-24**. Latest audited signal: **2026-04-24**.

1. Executive Summary

V17 Conservative is a weekly, rules-based SPY exposure model. It produces a Friday-after-close target exposure and an allocation across SPY, SPXL, SPXS, TLT, GLD, and SHY. It is designed for controlled paper/live testing, not discretionary day trading.

- Latest cached signal tested: **2026-04-24** | regime **NORMAL** | target allocation **SPY 100.0%**.
- Full-period result after 5 bps turnover costs: V17 CAGR **21.55%** vs SPY **14.58%**, Sharpe **1.171** vs **0.853**, max drawdown **-24.34%** vs **-33.72%**.
- 2021+ holdout: V17 CAGR **24.80%** vs SPY **14.55%**, Sharpe **1.409** vs **0.885**, max drawdown **-17.31%** vs **-24.50%**.
- Deployment audit outcome from this rerun: **16 PASS, 1 WARN, 0 FAIL**. The warning is data freshness because this environment could only test the uploaded offline cache.

Important: most of the historical edge comes from the V12 risk/crash engine. The V17 calm-bull sleeve improves upside, but V17 vs V12 is not strongly significant on Sharpe or drawdown by itself. Treat V17 as a conservative enhancement to V12, not a separate alpha engine.

2. Model Overview

Layer	Purpose	What it does
V12 base engine	Risk control + crash protection	Builds the core weekly signal: defensive 0.6x, normal 1.0x, levered recovery 1.2-1.7x, or crash-short -1.0x.
V17 conservative sleeve	Small upside boost	Only boosts V12 normal weeks to 1.15x when trend, momentum, realized volatility, and VIX all confirm a calm bullish regime.
Live tracker	Operational monitoring	Records each Friday signal once, reconstructs model returns, and compares against SPY and actual live results.

3. Inputs and Why They Were Chosen

Input group	Specific inputs	Reason
Tradeable assets	SPY, SPXL, SPXS	SPY is the benchmark exposure; SPXL/SPXS express limited long leverage or short exposure.
Defensive sleeve	TLT, GLD, SHY	Provides non-equity ballast during defensive/short regimes. The sleeve shifts toward GLD/SHY when stock-bond correlation becomes unfavorable.
Volatility stress	VIX, VIX3M, SKEW	Captures fear, short-term volatility pressure, term-structure stress, and tail-risk appetite.
Rates and curve	DGS10, T10Y2Y	Identifies rate-stress regimes and yield-curve changes that can suppress leverage or support recovery.
Credit stress	BAMLH0A0HYM2	High-yield option-adjusted spread is a market-based indicator of credit risk and stress.
Breadth	XLFX, XLK, XLE, XLY, XLP, XLU	Sector breadth confirms whether a post-drawdown recovery has broader participation.

4. Signal Logic and Allocation

The weekly signal is generated after Friday close. In the backtest and tracker, the signal becomes effective on the next trading day's close-to-close return as a one-bar delay. Actual Monday-open fills can differ from this proxy, so live execution prices must be recorded separately.

Signal	Regime	Target weights
-1.00x	CRASH_SHORT	SPXS 33%, TLT/GLD/SHY 67% total defensive sleeve
0.60x	DEFENSIVE	SPY 60%, TLT/GLD/SHY 40% defensive sleeve
1.00x	NORMAL	SPY 100%

Signal	Regime	Target weights
1.15x	CALM_BULL_BOOST	SPY 92.5%, SPXL 7.5%
1.20x	LEVERED_LONG_LOW	SPY 90%, SPXL 10%
1.50x	LEVERED_LONG_HIGH	SPY 75%, SPXL 25%
1.70x	POST_CRASH_BOOST	SPY 65%, SPXL 35%

Dollar scaling is direct: asset dollars = approved model capital x target weight. For a 1,000,000 pilot book and a 1.15x signal, allocate 925,000 to SPY and 75,000 to SPXL, creating approximately 1.15x net equity exposure.

5. Rerun Performance Results

Metric	SPY	V17	Difference
Full total return	950.2%	2815.3%	+1865.1%
Full CAGR	14.58%	21.55%	+6.97%
Full Sharpe	0.853	1.171	+0.318
Full max drawdown	-33.72%	-24.34%	+9.38 pp
2021+ total return	105.1%	222.9%	+117.7%
2021+ Sharpe	0.885	1.409	+0.523

6. Statistical and Robustness Checks

Check	Result from rerun	Interpretation
Block bootstrap vs SPY	Full Sharpe p=0.0033; holdout Sharpe p=0.0233; drawdown p approx 0.15-0.16	Return and Sharpe pass; drawdown advantage is directionally good but statistically noisy.
Random rebalance timing	0/300 random schedules beat V17 on return, Sharpe, Calmar, or max drawdown	The timing of exposures adds value versus random exposure scheduling.
Synthetic null markets	Sharpe/Calmar/maxDD p=0.0000; total return p=0.0267	Edge weakens when real regime structure is broken, supporting the regime-based thesis.
Deflated Sharpe	V17 probability 99.16% after assuming 80 tested variants	Sharpe remains strong after a multiple-testing haircut.
Rolling windows	V17 beat SPY in 100% of 148 rolling 5-year windows	Long-horizon consistency is strong across 2009-2026.
Parameter sensitivity	Small smoke grid: 100% beat V12 on holdout return/Calmar; 50% on Sharpe; 0% on maxDD	V17 boost improves upside, not drawdown versus V12.

7. Live Testing and Monitoring Plan

- Every Friday after US close: run the refreshed signal command, inspect latest_signal.json and V17_REPORT.md, then update the live tracker.
- Before execution: run the deployment audit. A FAIL blocks execution. A WARN requires manual review, usually data freshness or source-quality checks.
- Execution: place Monday-open trades according to target weights. Record actual fill price, quantity, commissions, slippage, and any failed/partial fills.
- Weekly review: compare actual live PnL versus SPY and versus the model tracker. Investigate if realized PnL differs materially from tracker PnL after accounting for Monday-open execution timing.
- Monthly review: run partial validation and review live Sharpe, drawdown, slippage, turnover, signal accuracy, and tracking error.
- Quarterly review: run deep validation and decide whether to continue the pilot, scale, freeze capital, or retire the model.

Live metric	Why it matters	Action trigger
Actual vs tracker return	Shows execution slippage and implementation quality	Investigate if gap exceeds expected bid/ask plus Monday-open timing effects.
Actual vs SPY return	Main benchmark comparison	Review if underperforming for several months, but do not overreact to short windows.
Live max drawdown	Main treasury risk measure	Escalate if drawdown exceeds model expectations or risk budget.
Signal ledger duplicates	Operational safety	Same-week rerun must update the existing row, not create a new trade instruction.
Data freshness	Prevents stale signals	Do not execute a live signal from stale cache data.
Turnover and cost	Validates cost assumptions	Stress if effective costs exceed modeled 5-15 bps.

8. Commands

Use case	Command
Weekly signal + tracker	py spy_v17_conservative.py --mode signal --refresh-data --live-track
Signal only	py spy_v17_conservative.py --mode signal --refresh-data
Offline test	py spy_v17_conservative.py --mode signal --offline --live-track
Deployment audit	py v17_deployment_audit.py --offline
Deployment audit with refresh	py v17_deployment_audit.py --refresh-data
Monthly validation	py spy_v17_conservative.py --mode full --refresh-data --n-bootstrap 300 --n-random-rebalance 300 --n-synthetic 75 --n-bayes-draws 10000 --sensitivity-grid smoke
Quarterly deep validation	py spy_v17_conservative.py --mode full --refresh-data --n-bootstrap 2000 --n-random-rebalance 1000 --n-synthetic 100 --n-bayes-draws 20000 --sensitivity-grid smoke

9. Controls Before Real Capital

- Use a formal pilot allocation and do not scale above the approved risk budget until live tracking has enough observations.
- Keep the code frozen during live testing. Any rule change creates a new model version and should restart the evaluation clock.
- Require human sign-off when signal changes into CRASH_SHORT, POST_CRASH_BOOST, or any new maximum exposure state.
- Maintain raw input data snapshots, output reports, signal ledgers, and fill records for auditability.
- Do not claim the backtest guarantees future performance. The model was validated on 2009-2026; a new macro regime can reduce or remove the edge.

10. Bottom Line

The model mechanics passed the rerun checks performed here. The strongest case is V17 versus SPY: higher historical return, higher Sharpe, lower drawdown, strong rolling-window consistency, and strong random-timing/null-market tests. The correct deployment decision is controlled paper/live pilot testing with strict tracking, not immediate unbounded treasury deployment. The model is suitable for a disciplined pilot if the Friday data is refreshed, the audit passes, and actual execution is recorded separately from the backtest tracker.